FastSurfer vs FreeSurfer concordance: what the evidence says for your MTLE-HS study

Use a single pipeline for all subjects — pipeline mixing is not acceptable for normative modeling. Published validation data show that FastSurfer and FreeSurfer agree well at the group level (test-retest ICC **0.92 vs 0.87** for cortical thickness), PubMed Central nih but between-pipeline agreement is only moderate (ICC ~0.62–0.87 depending on measure and population), PubMed Central with systematic offsets that would confound normative z-scores. Every published normative modeling study in epilepsy enforces strict pipeline consistency, no harmonization method has been validated for correcting pipeline differences, and your sample size (~40 controls) is too small to reliably estimate pipeline batch effects. For spectral shape descriptors (HKS/WKS/GPS), the risk is even greater because mesh construction differences between the two pipelines propagate directly into Laplace-Beltrami eigenpairs.

Cortical thickness agreement is good but not interchangeable

The original FastSurfer validation (Henschel et al., 2020, NeuroImage) established core concordance metrics against FreeSurfer v6.0 across five datasets (649 scans). Volumetric segmentation Dice was **89.08** for subcortical and **85.88** for cortical structures. arXiv nih Surface-based parcellation Dice against Mindboggle-101 manual labels reached **86.4–87.0** across hemispheres. Critically, no ROI fell below Dice **72.3**. arXiv nih Group separability testing on OASIS1 (AD vs CN) showed FastSurfer detecting cortical thinning with **smaller p-values** than FreeSurfer in the majority of structures, Dzne nih demonstrating that concordance does not sacrifice sensitivity. Deep-mi Helmholtz

However, the more clinically relevant question — direct between-pipeline agreement — tells a more cautious story. A 2025 pediatric validation (Zughayyar et al., Brain and Behavior; N=448, ages 4–18) comparing FastSurfer v2.3.3 to FreeSurfer v7.4 PubMed Central reported mean cortical thickness ICC of only **0.62** (95% CI: −0.07 to 0.87), PubMed Central overall volume DSC of **0.90** (CI: 0.79–0.95), and volume ICC of **0.87** (CI: 0.52–0.94). PubMed Central The wide confidence intervals reflect genuine regional variability in agreement. Part of this discrepancy stems from a training mismatch: FastSurfer was trained on FreeSurfer v6.0 labels, and FreeSurfer v7.x changed white matter and pial surface estimation algorithms. PubMed Central **No published study reports a mean absolute thickness difference in mm between the two pipelines**, nor has a formal Bland-Altman analysis been conducted — a notable gap. The pediatric data hint that FastSurfer may produce slightly thinner cortical thickness estimates relative to FreeSurfer v7.x normative charts, but this has not been formally quantified. PubMed Central

For scan-rescan reliability, FastSurfer consistently outperforms FreeSurfer. Deep-mi On OASIS1 test-retest data (20 subjects), FastSurfer achieved mean cortical thickness ICC of **0.92** versus

FreeSurfer's **0.87**, with FastSurfer superior in **30 of 31** DKT cortical regions. No FastSurfer region fell below ICC **0.78** (vs. **0.73** for FreeSurfer). PubMed Central Subcortical volume ICC was **0.99** for FastSurfer vs **0.97** for FreeSurfer across all 13 structures. PubMed Central nih The FastSurfer-LIT study (Estrada et al., 2025, Imaging Neuroscience) independently confirmed higher ICC for FastSurfer in pre/post-operative thickness comparisons. PubMed Central MIT Press

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Surface geometry differs in construction but converges in result

FastSurfer and FreeSurfer construct cortical surfaces through fundamentally different algorithmic paths. FastSurfer replaces FreeSurfer's `mri_tessellate` with **marching cubes** for initial mesh generation and replaces `mris_sphere` with a **one-shot spectral spherical embedding** using the first three Laplace-Beltrami eigenfunctions. PubMed Central +2 These differences produce meshes with distinct connectivity patterns — vertices are not in one-to-one correspondence between the two pipelines for the same subject. FastSurfer meshes have measurably higher triangle quality (quality index Q = **0.902** vs **0.888**, $p < 10^{-16}$) and **15% fewer topological defects** per hemisphere (23.1 vs 27.2 defects). nih

Average surface displacement between the two pipelines is **sub-voxel (<0.5 mm)** ResearchGate based on the FastCSR comparison study (Ren et al., 2022, Brain Informatics), ResearchGate with the white surface showing tighter agreement than the pial surface. No published study reports surface area or curvature ICC between pipelines — another significant gap. Known problem regions include the **temporal pole, insula, entorhinal cortex, and medial orbital frontal cortex**, which are challenging for all cortical reconstruction pipelines. FastSurfer explicitly masks lateral orbitofrontal regions during brainmask creation to prevent pial surface capture of the optic nerve. nih The pediatric validation noted a consistent unexplained **leftward asymmetry in the cingulate gyrus** specific to FastSurfer. PubMed Central

For parcellation, FastSurfer uses a fundamentally different approach than FreeSurfer: it maps DKT volume segmentation directly onto the reconstructed surface rather than using spherical atlas registration with `mris_ca_label`. PubMed Central +2 This produces surface parcellation Dice of **86–87** against manual labels. ResearchGate nih For Schaefer-200 or other fine-grained parcellations requiring `mri_surf2surf`, FastSurfer supports FreeSurfer's spherical registration (`mris_register` to `sphere.reg`) as an optional step, enabling cross-subject correspondence. PubMed Central nih However, **no Schaefer-specific validation exists**, and boundary misalignments from different surface geometries may introduce small errors at parcellation edges.

FreeSurfer 8.x represents a significant but well-characterized departure from 7.x

FreeSurfer 8.0.0 (released February 2025) integrates three deep learning components into the

default `recon-all` pipeline: **SynthSeg** replaces atlas-based Bayesian segmentation for initial volumetric labeling, **SynthStrip** replaces watershed-based skull stripping, and **SynthMorph** provides deep-learning-based nonlinear registration. (Free Surfer) Processing time drops from ~8–10 hours (v7.x) to **~2 hours** (v8.x). (Free Surfer) The release notes explicitly state: **"recon-all will have different output than version 7."** (Free Surfer)

The fundamental surface placement algorithms (`mris_place_surface`) remain unchanged, but the segmentation that initializes and guides surface placement is entirely different. FreeSurfer 8.x also introduces targeted DL-based corrections for historically problematic regions: improved white matter segmentation in the **entorhinal cortex** (where WM is often only ~0.5 mm thick), automatic DL segmentation of **major venous sinuses** and **MCA/dura artifacts** that previously caused pial surface errors, and corrections to white surface placement near **inferior frontal cortex/basal ganglia**. (Free Surfer) These changes mean the entorhinal cortex and temporal pole — regions directly relevant to MTLE-HS — will show different surface placement in v8.x compared to v7.x.

No peer-reviewed publication directly comparing FS 7.x vs 8.x cortical thickness has appeared as of March 2026. The FreeSurfer developers report that across several internal studies (aging, AD cross-sectional, AD longitudinal, schizophrenia), there were **"no major changes in the ability to detect effects"** and **"no interaction between effect and version."** (The Mail Archive) However, prior version comparison studies (Gronenschild et al., 2012; Bhagwat et al., 2024 bioRxiv) found between-version cortical thickness differences of **2.8 ± 1.3%** (FS 4.x vs 5.0) (PubMed) (PLOS) and showed that for some regions, between-version variability exceeded between-subject variability. (bioRxiv) (PLOS) The official FreeSurfer guidance remains unequivocal: **"Mixing versions is never a good idea."** (Free Surfer) (Free Surfer) Users can revert to pre-8.x behavior by setting `FS_V8_XOPTS=0`, though this eliminates the speed gains. (Free Surfer)

Spectral descriptors are theoretically robust but empirically untested across pipelines

The Laplace-Beltrami eigenvalue convergence rate for p-order finite elements is **$O(h^{2p})$**, where h is mesh element size (Reuter et al., 2009). (PubMed Central) For the cotangent-weight FEM on ~150K-vertex cortical meshes ($h \approx 1$ mm), this yields very small eigenvalue errors for the first 50–100 eigenvalues used in typical BrainPrint analyses. Lower eigenvalues capturing global shape are most stable; higher eigenvalues encoding local detail are progressively more sensitive to mesh perturbation. Formal convergence proofs (Dey, Ranjan & Wang, 2010) confirm that discrete Laplace-Beltrami spectra converge to continuous manifold spectra as mesh density increases. (Academia.edu)

Among spectral descriptors, **HKS is the most robust** to mesh perturbation because the exponential decay term $e^{-\lambda t}$ naturally suppresses high-frequency eigenvalue errors —

particularly at larger time scales capturing global shape features. WKS, using energy-localized log-normal filters without exponential damping, is **more sensitive** to specific eigenvalue perturbations. GPS, using scaled eigenfunctions directly ($1/\sqrt{\lambda}$ weighting), offers intermediate robustness. A comprehensive 2024 IEEE TVCG benchmark of nine spectral descriptors across ten datasets confirmed that HKS is generally robust to mesh discretization and moderate geometric noise, while WKS and GPS show greater sensitivity. ResearchGate

The critical caveat: **no published study has directly compared LBO eigenpairs or spectral descriptors computed on FastSurfer versus FreeSurfer surfaces for the same subjects.** This remains a clear gap in the literature. Indirect evidence is encouraging — BrainPrint achieved **99.9% subject identification accuracy** across 3,000 FreeSurfer-processed scans Mit (Wachinger et al., 2015), suggesting inter-subject spectral differences vastly exceed within-subject noise. ScienceDirect The BrainPrint Python package now explicitly supports both FreeSurfer and FastSurfer segmentations as input. Helmholtz However, systematic between-pipeline spectral differences would be confounded with group membership in a normative modeling framework, regardless of their magnitude relative to biological signal.

An important distinction exists between **eigenvalue-based analyses** (ShapeDNA/BrainPrint) and **vertex-wise spectral descriptors** (HKS/WKS/GPS at each point). Eigenvalues depend only on intrinsic geometry, not mesh connectivity, making them inherently more robust to pipeline differences. Deep-mi Vertex-wise descriptors depend on eigenfunctions, which are affected by mesh connectivity — and since FastSurfer and FreeSurfer vertices are not in correspondence, vertex-wise comparison requires resampling to fsaverage, introducing additional interpolation error.

Normative modeling demands pipeline consistency without exception

The literature is unanimous: **every published normative modeling study in epilepsy enforces strict processing pipeline consistency.** The ENIGMA-Epilepsy consortium (Caciagli et al., 2020; 804 MTLE-HS patients, 1,625 controls, 25 centers) mandates FreeSurfer across all sites. The PCNtoolkit protocol (Rutherford et al., 2022, Nature Protocols) requires FreeSurfer recon-all for all inputs. PubMed Central +2 CentileBrain (Ge et al., 2024, Lancet Digital Health; N=37,407) explicitly tracks FreeSurfer version as a covariate. PubMed Central Brain MoNoCle (Little et al., 2025), validated specifically on TLE data, uses FreeSurfer throughout. MIT Press A practical guide for normative modeling in epilepsy (Taylor, Wang & Leiberg, 2025) processed all subjects (23 mTLE patients, 70 controls) with FreeSurfer recon-all using the DK atlas. arXiv arXiv

No harmonization method has been validated for correcting processing pipeline differences. ComBat (Fortin et al., 2018) was designed for scanner/site batch effects PubMed and assumes a common processing pipeline — a bioRxiv study explicitly concluded that **"the same processing pipeline should be used while using ComBat."** bioRxiv CovBat (Chen et al., 2022) extends this

to covariance structure (PubMed) but carries the same limitation. Hierarchical Bayesian approaches in PCNtoolkit model site as a random effect but would need to treat "pipeline" as a batch variable with only two levels (FastSurfer vs FreeSurfer) at N ≈ 40 per group — statistically underpowered and unreliable. A travelling-subjects validation study showed that harmonization can even **introduce bias where none existed** when applied inappropriately. (ISMRM)

The effect sizes at stake reinforce the risk. MTLE-HS produces moderate cortical thinning (Cohen's d ⩾ 0.5 in temporal regions per ENIGMA-Epilepsy), while pipeline-related systematic biases in cortical thickness range from **2–15%** depending on region and pipeline pair (PLOS) (Seiger et al., 2018; (PubMed Central) Gronenschild et al., 2012). (PLOS) These magnitudes overlap directly, meaning pipeline artifacts could mimic, obscure, or amplify disease effects in normative z-scores.

Practical recommendations for your MTLE-HS study design

For a study with ~40 healthy controls and ~46 MTLE-HS patients using cortical thickness, curvature, sulcal depth, and spectral shape descriptors for GP normative modeling, the evidence supports these specific recommendations:

- **Process all subjects with a single pipeline and version.** If you plan to use pre-trained normative models (PCNtoolkit, CentileBrain, Brain MoNoCle), use FreeSurfer to match the training data — ideally the version the model was trained on (typically v6.0 or v7.x). If building your own GP model from local controls, either FreeSurfer or FastSurfer is acceptable, but all subjects must use the same one.
- **FastSurfer is a valid choice if used consistently.** Its higher test-retest reliability (ICC 0.92 vs 0.87), (PubMed Central) faster processing (~1h vs ~8h), (PubMed Central) and fewer topological defects (nih) make it technically attractive. (GitHub) (Deep-mi) However, note that all major normative reference datasets were built with FreeSurfer, (eLife) limiting interoperability with pre-trained models.
- **For spectral descriptors specifically, prefer eigenvalue-based measures (ShapeDNA/BrainPrint) over vertex-wise HKS/WKS/GPS** if any uncertainty about mesh consistency exists. If using vertex-wise spectral descriptors, restrict analysis to larger time scales (HKS) or lower frequencies where robustness to mesh perturbation is strongest. Limit eigenfunction analysis to the first **20–50 modes** for maximum stability.
- **If data has already been processed with different pipelines, reprocess** one cohort to match the other rather than attempting post-hoc harmonization. The reprocessing cost (~1h per subject with FastSurfer, ~8h with FreeSurfer v7.x, ~2h with FreeSurfer v8.x) is modest compared to the risk of confounded results. (GitHub) (nih)

- **Do not mix FreeSurfer versions** (e.g., 7.x and 8.x) within a study. Free Surfer Free Surfer FreeSurfer 8.x changes to entorhinal cortex and temporal pole surface placement are particularly relevant to MTLE-HS, where these regions carry diagnostic signal. If starting fresh, FreeSurfer 7.4.1 remains well-characterized and widely used; v8.x offers speed gains but lacks published cross-version thickness comparisons.
- **Report pipeline details exhaustively**: software version, exact command-line flags, any manual edits, quality control criteria (Euler number thresholds, ENIGMA QC protocols). With N = 40 controls, you have adequate but not generous calibration data — the practical guide suggests N ⩾ 30 provides substantial improvement over smaller samples, arXiv and site adaptation using your local controls is essential when leveraging pre-trained models. Emergent Mind

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Conclusion

The concordance evidence paints a clear picture: FastSurfer and FreeSurfer produce **highly correlated but not interchangeable** outputs. Between-pipeline cortical thickness ICC ranges from moderate (0.62) to good (0.87) depending on population and FreeSurfer version, with systematic offsets that remain uncharacterized in mm. PubMed Central Test-retest reliability favors FastSurfer. PubMed Central nih Surface geometry agrees to sub-voxel precision, ResearchGate but mesh construction differences propagate into spectral analysis through different vertex connectivity, triangle quality, and eigenfunction patterns — effects that have never been directly quantified. FreeSurfer 8.x introduces its own discontinuity with v7.x through SynthSeg-initialized segmentation and targeted surface placement corrections in temporal regions critical to epilepsy research. Free Surfer

The most important finding may be the gap that doesn't exist: no published study, consortium guideline, or methodological framework endorses mixing processing pipelines within a normative modeling analysis. The recommendation is not cautious hedging — it reflects a structural limitation. Pipeline differences introduce systematic, regionally heterogeneous biases that cannot be reliably separated from biological signal at sample sizes typical of clinical neuroimaging studies. For your MTLE-HS study, choose one pipeline, apply it consistently, and report it transparently.